

A Universe in a Bottle — The Eleven-Paper Series

A peer-review-style write-up of the BCC quantum-cellular-automaton (QCA) model of physics. Each paper is self-contained and replication-grade: it states the construction, gives the derivations, quotes the verification residuals, names the underlying findings (findings/F*.md) and modules (ca-simulation/ca_*.py), and lists external references.

Issued 2026-06-08. Supersedes the ten-paper draft in deprecated/papers-v1/ (the matter sector is now split into three papers — fermions, leptons, baryons — and the dielectric black hole has its own paper). Canonical SI cell aligned to Finding F107 throughout: $a = \sqrt{8\pi} 3^{1/4} \ell_P = 6.5978 \ell_P = 1.0664 \times 10^{-34} \text{ m}$.

Reading order

#	Paper	What it establishes
I	Base Structure	BCC Weyl QCA substrate; speed of light as a rotation rate ($c_{\text{lat}} = 1/\sqrt{3}$); Maxwell as a linearised rotation; the spherical-Pythagorean mass shell.
II	The Photon	The photon as a bound pair of two spin- $1/2$ lattice quanta; massless, luminal, transverse, exactly non-birefringent; supersedes the chiral σ -bilinear.
III	Higgs-Free Mass	Chiral- $SU(2)$ complex-mass step (no Higgs); the $SU(2)_L$ Ward identity; the NJL constituent-mass scale; the Koide relation as cubic equipartition.
IV	Electromagnetism	Maxwell from the rotation rule; $U(1)$ minimal coupling forces the paired photon; Aharonov-Bohm + sourced curl end-to-end; charge quantisation and anomaly freedom.

#	Paper	What it establishes
V	Strong Force	Dynamical $SU(3)$ gluons (even propagator, forced); confinement from exact 2D area law $\rightarrow$ colour-dielectric dual superconductor $\rightarrow$ gauge Monte-Carlo; centre-phase closure.
VI	Weak Force	Chiral $SU(2)_L \times U(1)_Y$ without a Higgs; rank-1 Stueckelberg masses; the derived Weinberg angle ($\sin^2 \theta_W = \frac{1}{4}$); end-to-end β -decay.
VII	Gravity	Gravity as an impedance-matched lattice dielectric; $K = e^{2GM/rc^2}$; PPN $\beta = \gamma = 1$; the structural Newton constant $G = a^2 c^3 / (8\pi \sqrt{3} \hbar)$.
VIII	Dielectric Black Hole	The exponential metric: horizon-free; throat e , photon sphere $2\sqrt{e}$, shadow $2e$ (+4.63% vs Schwarzschild); no Hawking glow; GW echoes. (Reviews F114.)
IX	Fermion Sector	Weyl/Dirac quanta; chirality-helicity correspondence; exactly three generations from O_h (no fourth); anomaly-free first generation.
X	Lepton Sector	Higgs-free hypercharge; the Koide 45° equipartition as an EM-selected critical point; the Higgs-free see-saw neutrino.

#	Paper	What it establishes
XI	Baryon Sector	The colour-singlet proton; centre-phase closure (baryon mass = field energy); the dynamical pion; the deuteron bound by the tensor force; the derived NN repulsive core.

Headline verified numbers

- $c_{\text{lat}} = 1/\sqrt{3} = 0.5773503$ (rotation-rate slope, axis and body diagonal)
- $\sin^2 \theta_W = \frac{1}{4} \Rightarrow m_Z/m_W = 2/\sqrt{3} = 1.1547$ (zero-parameter; 1.77% from PDG)
- Koide $Q = 0.6666605$ (0.91σ from $\frac{2}{3}$); $m_\tau^{\text{pred}} = 1776.97$ MeV (6.1×10^{-5})
- $a/\ell_P = \sqrt{8\pi} 3^{1/4} = 6.5978$; $G = 6.6743 \times 10^{-11}$ SI (3×10^{-8} , CODATA round-off)
- Black-hole shadow ratio $2e/3\sqrt{3} = 1.0463$ (+4.63%); throat e , photon sphere $2\sqrt{e} = 3.297$, shadow $2e = 5.437$ (units GM/c^2)
- Deuteron $E_b = 2.224$ MeV, $\kappa = 0.2316$ fm $^{-1}$; baryon constituent-mass fraction 0.96% of m_p

Standing design decisions (deviations from the Standard Model)

1. Speed of light = rotation rate, not phase velocity (F26).
2. Mass from chiral- $SU(2)$ complex coupling, not a Higgs field; hypercharge on the same gauge field (F27/F41).
3. Photon = bound pair of two spin- $1/2$ quanta, not a fundamental spin-1 boson (F69).
4. Gravity = a single impedance-matched lattice dielectric, $K = e^{2GM/rc^2}$ (F64).

See `key-decisions.md` and `findings/` for the full record.